

Module

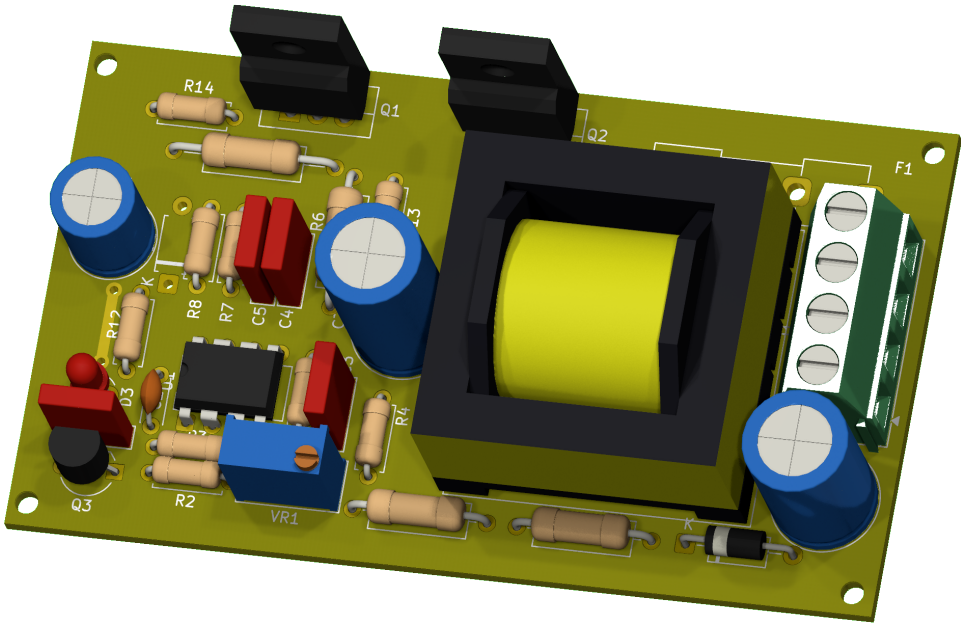
Variant: CHECKED

Rev 1.1.0+ (Unreleased)

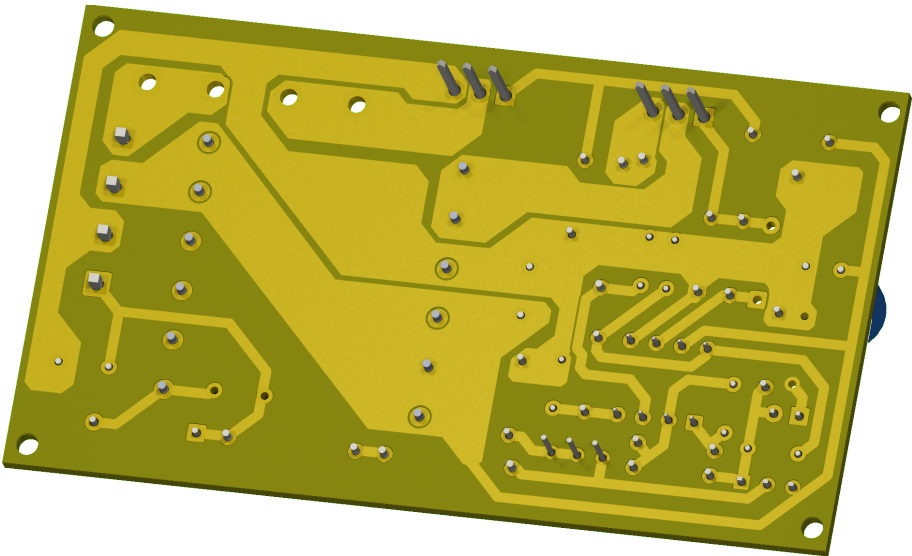
2026-01-17

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TOP VIEW



BOTTOM VIEW



DRAFT - Early concept stage, schematic drafted.

PRELIMINARY - Minor schematic changes are expected, ready for layout planning.

CHECKED - DRC completed, schematic and layout ready for review.

RELEASED - Design ready for manufacturing.

NOTE STYLING CONVENTION

DESIGN NOTE:

Example text for informational design notes.

DESIGN NOTE:

Example text for debug notes.

DESIGN NOTE:

Example text for cautionary design notes.

DESIGN NOTE:

Example text for critical design notes.

LAYOUT NOTE:

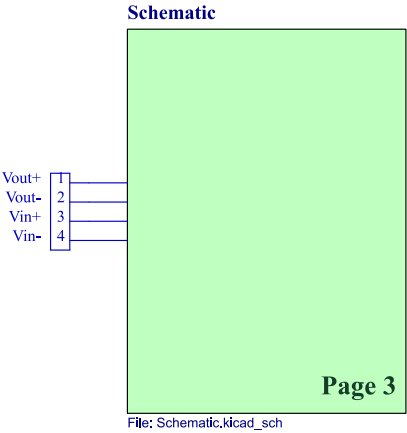
Example text for critical layout guidelines.

Date: 17-Jan-2026

	Comments:	Company: FR	Variant: CHECKED	Git Hash: 25e1572
	Board Name: Module		Project Name: Step-up module 12:450 V	
	Sheet Title:	File Name: StepUp_module_12to450V.kicad_sch	Designer: FR	Date: 2025-12-20
	Sheet Path: /	Reviewer: NA	Size: A3	Revision: 1.1.0+ (Unreleased)
				Sheet: 1 of 4

[2] Block Diagram

MODULE INTERFACE:
Vin: 9 V to 24 V according
to AliExpress product page
Vout: 200 V to 450 V,
adjustable through VR1



	Comments: AliExpress module example (link)		Company: FR		Variant: CHECKED		Git Hash: 25e1572	
			Board Name: Module			Project Name: Step-up module 12:450 V		
	Sheet Title: Block Diagram		File Name: Block Diagram.kicad_sch		Designer: FR		Date: 2025-12-20	
	Revision: 1.1.0+ (Unreleased)							
Sheet Path: /Block Diagram/					Reviewer: NA		Size: A4	
							Sheet: 2 of 4	

[3] Schematic

Estimated specs for unidentified / custom parts:

- F1: ~5 A
- D4: ~500 mA x 600 V
- R9: ~13.7 mΩ (110 mV @ 8.04 A)
- TR1: ~1 : 19 turns ratio, 3.54 μH : 1.45 mH

See "Supporting Docs\Measurements.xlsx" for details

	Comments:	Company: FR		Variant: CHECKED	Git Hash: 25e1572
		Board Name: Module		Project Name: Step-up module 12:450 V	
	Sheet Title: Schematic	File Name: Schematic.kicad_sch	Designer: FR	Date: 2025-12-20	Revision: 1.1.0+ (Unreleased)
	Sheet Path: /Block Diagram/Schematic/		Reviewer: NA	Size: A4	Sheet: 3 of 4

	Comments:	Company:		Variant:	Git Hash:
		FR		CHECKED	25e1572
	Board Name:		Project Name:		
	Module		Step-up module 12:450 V		
Sheet Title:	File Name:	Designer:	Date:	Revision:	
Schematic	Schematic.kicad_sch	FR	2025-12-20	1.1.0+ (Unreleased)	
Sheet Path:		Reviewer:	Size:	Sheet:	
/Block Diagram/Schematic/		NA	A4	3 of 4	

[4] Revision History

Version 1.0.0 - 2025-12-19

First issue

Schematic capture complete, except for parts:

- not populated (D2, C9);
- with unreadable or absent markings (F1, D4, R9, TR1).

CI failed on release - do not use.

Version 1.0.1 - 2025-12-21

Changed

Changelog updates only - issue with missing changelog on revision 1.0.0

Version 1.1.0 - 2026-01-16

Changed

Replaced UC2843A by UC3845 - output signal is active only once every two cycles, indicating the IC matches UCx845 functionality.

Added

Measurement data (waveforms, transformer characteristics). See "Supporting docs"

	Comments:	Company: FR		Variant: CHECKED	Git Hash: 25e1572
		Board Name: Module		Project Name: Step-up module 12:450 V	
	Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: FR	Date: 2025-12-20	Revision: 1.1.0+ (Unreleased)
	Sheet Path: /Revision History/		Reviewer: NA	Size: A4	Sheet: 4 of 4